3D METALLIC BONDING

SPECIFICATION REFERENCE
3.20 3.21 3.22

LEARNING OBJECTIVES

- Understand that metals consist of giant lattices of metal ions (cations) in a sea of delocalised electrons.
- Know that metallic bonding is the strong electrostatic attraction between cations and the delocalised electrons.
- Be able to use the above two models to interpret simple properties of metals, e.g. electrical conductivity and high melting temperature.

THE NATURE OF METALLIC BONDING

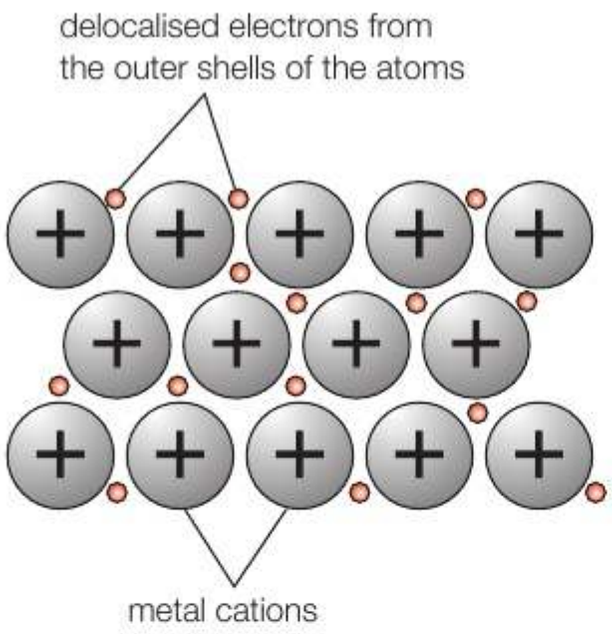
Metals typically have the following physical properties:

- high melting temperatures
- good electrical conductivity
- good thermal conductivity
- malleability
- ductility.

Any theory of the way that the atoms in a metal are bonded together must explain the above properties.

Metals typically have one, two or three electrons in the outer shell of their atoms and have low ionisation energies. The electrical conductivity of a metal generally increases as the number of outer-shell electrons increases.

Since electrical conductivity depends on the presence of mobile carriers of electric charge, we can build a picture of a metal as consisting of an array of atoms with at least some of their outer-shell electrons removed and free to move throughout the structure. These **delocalised electrons** are largely responsible for the characteristic properties of metals.



▲ **fig A** Diagram showing the particles present in a metal.

EXAM HINT

If asked to draw a diagram of a metallic bond in an exam, make sure you include a key showing ions and electrons.

The electrons are said to be delocalised since they are free to move throughout the structure and are not confined, i.e. localised, between any pair of cations.

There are electrostatic forces of attraction between the cations and the delocalised electrons. This is known as **metallic bonding**.

EXPLAINING THE PHYSICAL PROPERTIES OF METALS

MELTING TEMPERATURE

In order to melt a metal, it is necessary to partially overcome the forces of attraction between the cations and the delocalised electrons to such an extent that the cations are free to move around the structure. Metals have a giant lattice structure where many of these forces must be overcome. The energy required to do this is usually very large, so the melting temperatures are typically high.

The number of delocalised electrons per cation plays a part in determining the melting temperature of a metal.

- Group 1 metals have low melting temperatures.
- Group 2 metals have higher melting temperatures.
- Metals in the d-block typically have high melting temperatures because they have more delocalised electrons per cation.

Another factor that affects the melting temperature is the charge-to-radius ratio of the cation. The greater the charge-to-radius ratio, the stronger the attraction for the delocalised electrons. Therefore, for two cations of the same charge, the smaller cation will attract the delocalised electrons more strongly. This is why, for example, the melting temperature of lithium is greater than that of sodium.

ELECTRICAL CONDUCTIVITY

When a potential difference is applied across the ends of a metal, the delocalised electrons will be attracted to, and move towards, the positive terminal of the cell. This flow of electrons constitutes an electric current.

THERMAL CONDUCTIVITY

Two factors contribute to the ability of metals to transfer heat energy.

1 The free-moving delocalised electrons pass kinetic energy along the metal.

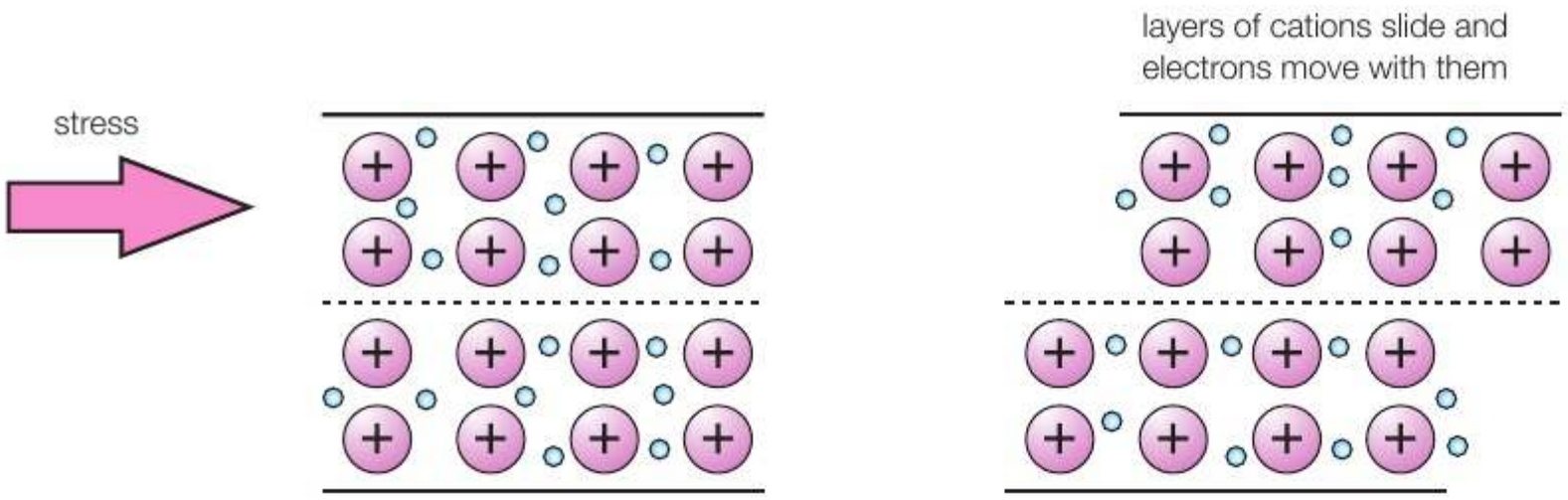
2 The cations are closely packed and pass kinetic energy from one cation to another.

The conduction by the delocalised electrons is by far the more significant of the two factors.

MALLEABILITY AND DUCTILITY

Metals can be hammered or pressed into different shapes (malleability). They can also be drawn into a wire (ductility). Both of these properties depend on the ability of the delocalised electrons and the cations to move throughout the structure of the metal.

When a stress is applied to a metal, the layers of cations may slide over one another (**fig B**).



▲ **fig B** The effect of stress on a metal.

However, because the delocalised electrons are free moving, they move with the cations and prevent strong forces of repulsion forming between the cations in one layer and the cations in another layer.

DID YOU KNOW?

The argument relating charge-to-radius ratio of the cation to the melting temperature is over-simplified. The way the ions are arranged in the lattice also affects the melting temperature. This is why there is no regular trend in the melting temperatures of the Group 2 metals. The way in which the ions are arranged in the lattice changes down the group from beryllium to radium. This is not included in this book and is not required for IAS or IAL studies.

CHECKPOINT

1. Explain what is meant by the term 'metallic bonding'.

2. Suggest why the melting temperatures and electrical conductivities of sodium, magnesium and aluminium are in the order Na < Mg < Al.

SUBJECT VOCABULARY

delocalised electrons electrons that are not associated with any single atom or any single covalent bond

metallic bonding the electrostatic force of attraction between the metal cations and delocalised electrons